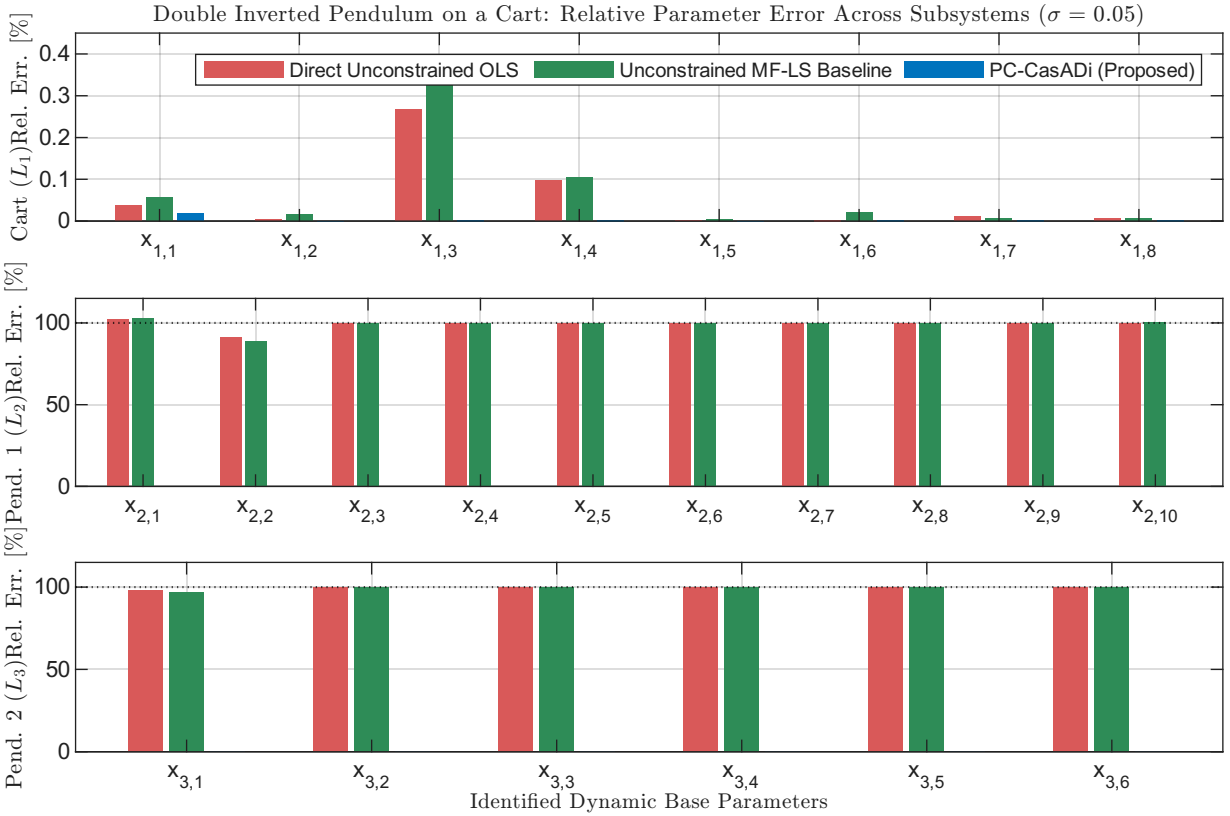
Double Inverted Pendulum on a Cart: Relative Parameter Error Across Subsystems ($\sigma = 0.05$)
Direct Unconstrained OLS
Unconstrained MF-LS Baseline
PC-CasADi (Proposed)
Cart (L_1)Rel. Err. [%]
Pend. 1 (L_2)Rel. Err. [%]
Pend. 2 (L_3)Rel. Err. [%]
0
0.1
0.2
0.3
0.4
50
100
$x_{1,1}$
$x_{1,2}$
$x_{1,3}$
$x_{1,4}$
$x_{1,5}$
$x_{1,6}$
$x_{1,7}$
$x_{1,8}$
$x_{2,1}$
$x_{2,2}$
$x_{2,3}$
$x_{2,4}$
$x_{2,5}$
$x_{2,6}$
$x_{2,7}$
$x_{2,8}$
$x_{2,9}$
$x_{2,10}$
$x_{3,1}$
$x_{3,2}$
$x_{3,3}$
$x_{3,4}$
$x_{3,5}$
$x_{3,6}$
Identified Dynamic Base Parameters